James Bray

Aerospace Engineering Student, MEng, Final year

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Professional Profile

A versatile Aerospace Engineering student (on track for 1st Class MEng) with a background spanning software development, systems integration, and advanced automation. Specialises in creating end-to-end solutions for complex engineering problems, from designing modular UAV avionics and state space controllers to building custom full-stack software and digital twins. Proven ability to adapt quickly to new technical challenges and deliver high-impact results in both software and hardware arenas.

Education & Experience

2022/09 - present

Aerospace Engineering Degree MEng *University of Sheffield, predicted first-class*
Four-year course with a year in industry, currently in final year

Relevant modules:

- Dissertation: Optimising Intelligent Swarm Exploration for Large Environments
- VTOL quad plane: Took on 2 roles, leading all avionics aspects of the quad plane
- Machine Intelligence: Utilised advanced techniques in data processing and modelling
- Rapid Control Prototyping: Designed and optimised a controller for a helicopter system
- See all other modules and grades achieved on my website

2025/09 - 2025/12

UAV Swarm Engineer *XPRIZE AURA, team member*
XPRIZE Wildfire is an international competition to develop automated wildfire solutions

- Developed a full-stack drone navigation system for swarm management and coordination
- Communicated methodology and coordinated progress in cross-functional teams
- Built a 3D swarm simulation environment for fire detection computer vision testing
- Contributed to a paper outlining the novel optimisation methodology I developed
- Unfortunately, the project was discontinued as we didn't proceed to the finals

2024/07 - 2024/10

Assistant Project Engineer *The University of Sheffield AMRC, summer placement*
The AMRC is a leading research hub for advanced manufacturing, specialising in aerospace

- Designed and manufactured a £4000 partial discharge testing cell for 20kV motor coils
- Co-ordinated and liaised with 6 suppliers to manufacture components for construction
- Managed project deliverables, mitigated risks and produced supporting documentation
- Used Ansys FEA to evaluate the feasibility of geometries in slinky stator manufacture
- Utilised Ansys' design optimisation for the pneumatic end effector of a sorting robot
- Planned the automated disassembly of a Siemens servo motor to extract valuable parts

2021/06 - 2021/07

Catalogue Executive *MinsterFB, temporary full-time*
MinsterFB is an end-to-end agency helping brands grow on the Amazon platform

- Created and optimised listings for maximum engagement and profitability
- Used Excel to make an automated product name generator based on supplier data
- Automated multipack image generation with Python, saving ~30 minutes per product.
- Gained a good understanding of SEO for Amazon listings to help maximise conversions
- Adapted to working in a fast-paced environment both independently and in virtual teams

Highlighted Projects

2024/10 - present **Project Sunride - Avionics (propulsion)** *Extra-Curricular University Project*
Student-led team, building rockets and rocket engines with the end goal of reaching space

- Designing avionic systems for the university's first liquid rocket engine test stand
- Utilising LabJack DAQ systems to interface sensors and actuators for precise control
- Built a SIMULINK fluid model with PID control to regulate fuel and ox flow to engines
- Implementing fault detection and safety protocols for the rocket engine test procedures
- Exploring engine gimbaling systems on the test stand to control thrust direction

2024/02 - 2024/05 **Control Lead - Quadcopter Design, Build, and Test** *University Project*
Student teams compete to win a contract for drone manufacturing with a fictitious company

- Won the contract for the lightest flying drone out of a cohort of 31 teams
- Led control team, optimising flight behaviour through PID controller BetaFlight
- Used Google Sheets to analyse and predict motor performance with different components
- Modelled a digital twin in Fusion 360 to optimise the strength-to-weight ratio using FEA
- Manufactured a drone using laser-cut or 3D-printed parts, and soldered electronics

2022/10 - 2023/06 **UKSEDS Rocketry Competition** *Extra-Curricular University Project*
Design team member in UKSEDS, where teams design and build model rockets.

- Designed and 3D printed a whistling rocket nosecone using Fusion 360 and Prusa
- Modelled the rocket performance using OpenRocket to ensure maximum apogee
- Handled subteam conflicts regarding rocket dimensions and features
- Utilised MATLAB to perform Fast Fourier Transform (FFT) on launch audio data
- Evaluated the Doppler effect to compare with the acceleration data in the telemetry

2022/09 - 2022/10 **Self-Sustaining TurboJet Engine** *Personal Project*
Recycled a car turbo into a turbojet engine using tools in my garage.

- Researched existing designs and systems to inspire my sketches and prototypes
- Used Fusion 360 to model the assembly of the turbo and the combustion chamber
- Powered and coupled a 20,000 rpm starter motor to get the engine to operating speeds
- Manufactured a combustion chamber using tools like a pillar drill and an angle grinder
- Practised safety measures and mitigated hazards (high voltages, high temperatures)

2021/06 - 2021/07 **Topology Optimisation with Finite Element Analysis** *A-Level Coursework*
Developed an application for FEA and Topology Optimisation in 2D

- Designed and built UI/UX interfaces for my application in the Unity game engine
- Programmed the backend processing using C# and Math.NET libraries (for linear algebra)
- Coded algorithms like FEA, Marching Squares, Filtering, Binary Search and Serialisation
- Used project planning methods like Flowcharts, Gantt Charts, DFDs and Class Diagrams
- Created an extensive document that outlines and justifies every step of the development

Proficiencies

A summary of my current skillsets demonstrated in my curricular and extra-curricular experiences.

Technical Skills

- Simulation & Digital Twins
- System Design and Analysis
- Automation and AI
- Codebase development
- Documentation

Soft Skills

- Effective communication
- Analytical thinking
- Leadership and Collaboration
- Attention to detail
- Quick learning aptitude

Software

- AI Coding Tools
- MATLAB and Simulink
- Python, C, C#
- Unity & Blender
- Google & Microsoft Suites